

MetaKV supplementary methods and evidence

This supplement describes the frozen metadata and replay methods, documents execution provenance, and reports the existing packed-kernel sensitivity measurements. The measurements support platform-dependent kernel cost; they do not establish end-to-end serving performance or hardware fault incidence.

S1 Metadata and replay methods

Each group contains 32 signed INT4 values and one positive scale. HA-FBMS stores a coarse thermometer field and a binary fine field in 32 metadata bits; its decoder uses popcount rather than requiring a valid thermometer prefix after a fault. The manuscript equations specify the frozen codec. Model-specific log-scale intervals are $[-13, 3]$ for Llama-3.1-8B, $[-11, 5]$ for Qwen3-14B, and $[-13, 5]$ for Qwen3-32B, with zero observed clipping. The corresponding scale populations contain 276,480, 317,440, and 507,904 groups. The original full-cache replay performs one next-token step; the AMD P8 extension follows the same frozen logical locations for 32 steps. Selected metadata faults are conditional logical tests, not samples of a physical fault process.

S2 Deployment protocol

Correctness is checked before timing. Workloads contain 1,048,576 or 4,194,304 groups, with 32 values per group. Each configuration uses 30 warmup launches, 15 paired blocks, 100 launches per block, and 10,000 bootstrap resamples. The reported percentage is the median of paired block-level mapped-versus-FP32 differences, not a ratio of independent aggregate medians. Confidence intervals are the archived bootstrap intervals, without new significance tests. Each execution-platform summary contains eight configuration/workload rows and 120 paired block rows. Table S1 gives the post-freeze Linux sensitivity; Table S2 reports the largest workload across execution platforms.

S3 Linux RTX 5080 sensitivity

Execution platform is RTX5080_LINUX_H800_LIKE; execution GPU is NVIDIA GeForce RTX 5080. Runtime records identify driver 580.82.07, Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, and Triton 3.7.1. Ubuntu 22.04.5 and kernel 5.15.0-78 are from the supplied execution record. The same physical GPU is author-reported; the logs verify the model, not a shared serial identifier. This is not a strict OS-only A/B comparison. The maximum absolute mapped-versus-FP32 median is 0.168554942360% at 4,194,304 groups and 134,217,728 values. Negative timing differences do not imply acceleration. Three of the four intervals exclude zero, despite very small magnitudes; this is not a universal zero-overhead result.

Table S1. Linux RTX 5080 sensitivity at the largest workload

Representation	Bc/Bf	Median (%)	95% CI (%)
REP_0008	14/18	+0.072469	[+0.020975, +0.128995]
REP_0025	19/13	+0.168555	[+0.149584, +0.239737]
REP_0026	22/10	-0.089114	[-0.105037, -0.066795]
REP_0027	26/6	-0.013039	[-0.016849, +0.010451]

Warmup 30; paired blocks 15; launches per block 100; bootstrap resamples 10,000. Kernel microbenchmark only. The H800-like software stack on RTX 5080 does not reproduce the larger original H800 overhead; architecture-only causation is not established.

S4 Cross-platform evidence

Execution labels identify the measured hardware and runtime, while representation-source labels identify configuration provenance. H800_CUDA or AMD_ROCm in a representation row must not be interpreted as its execution GPU. Table S2 uses the original passing H800 freeze, not the failed grouped-decoder R6 diagnostic. AMD is reported as gfx942-class hardware. Missing historical runtime fields remain unverified rather than inferred.

Table S2. RTX5080_WINDOWS

Representation	Median (%)	95% CI (%)	CI crosses zero
REP_0008	-1.022627	[-2.186483, +1.747240]	True
REP_0025	-0.224026	[-1.063780, +0.512331]	True
REP_0026	-0.171614	[-0.661034, +0.962437]	True
REP_0027	-0.013033	[-0.724311, +0.299826]	True

Table S2 continued. RTX5080_LINUX_H800_LIKE

Representation	Median (%)	95% CI (%)	CI crosses zero
REP_0008	+0.072469	[+0.020975, +0.128995]	False
REP_0025	+0.168555	[+0.149584, +0.239737]	False
REP_0026	-0.089114	[-0.105037, -0.066795]	False
REP_0027	-0.013039	[-0.016849, +0.010451]	True

Table S2 continued. RTX4060TI_WINDOWS

Representation	Median (%)	95% CI (%)	CI crosses zero
REP_0008	+0.005121	[-0.378260, +0.429121]	True
REP_0025	-0.019257	[-0.306929, +0.485951]	True
REP_0026	+0.159240	[-0.403659, +0.471145]	True
REP_0027	+0.044226	[-1.011308, +0.430099]	True

Table S2 continued. AMD_GFX942

Representation	Median (%)	95% CI (%)	CI crosses zero
REP_0008	+165.067203	[+165.002506, +165.075680]	False
REP_0025	+162.554752	[+162.515095, +162.588667]	False
REP_0026	+156.963005	[+156.940219, +157.012195]	False
REP_0027	+154.973863	[+154.850956, +155.074009]	False

Table S2 continued. H800_CUDA

Representation	Median (%)	95% CI (%)	CI crosses zero
REP_0008	+16.783045	[+16.703419, +16.849694]	False
REP_0025	+12.550879	[+12.538873, +12.558011]	False
REP_0026	+10.920462	[+10.872689, +10.995703]	False
REP_0027	+1.640163	[+1.618926, +1.663246]	False

S5 Full-cache distributional evidence

Table S3 reports existing corrected Binary32-Z P6B2 global statistics. All statistics were checked against the 96 selected target rows per model. A zero median does not imply absence of large outliers. These are one-step localized injections on a full-cache INT4 background, not long-sequence generation or randomized device-level reliability estimates.

Table S3. Corrected Binary32-Z target-level KL distributions

Model / targets	Median KL	p95 KL	Maximum KL	Top-1 changes
LLAMA31_8B / 96	0.0	0.0015680934302508 83	0.0205116234719753 27	0
QWEN3_14B / 96	0.0	0.0004185554571449 7566	11.833429336547852	2

S6 AMD Qwen3-32B multi-step replay

The P8 formal run used eight frozen prompts and 24 frozen logical target locations on AMD Instinct MI308X. For each target, FP32, Binary32-Z, and HA-FBMS were tested, giving 72 cases. A single encoding-specific metadata bit was injected once into the initial full-cache representation. Targets were not reselected. Clean metadata was reconstructed locally on AMD at the frozen logical locations; the procedure is not an exact cross-backend clean-state reproduction. Teacher-forced and free-running greedy trajectories were evaluated at steps 1, 8, 16, and 32.

Across the 24 locations, the maximum, mean, median, and 95th-percentile absolute log-scale reconstruction errors were 0.0109711178210, 0.00262167446419, 0, and 0.0106799379959. Eighteen locations were within 1e-3 and 1e-2, and all 24 were within 5e-2. These values document frozen-location transfer rather than bit-identical metadata words across backends.

Table S4. AMD Qwen3-32B multi-step replay

Encoding	Validity	Diverged by steps 1 / 8 / 16 / 32	Step-32 summary
FP32	0 valid; 24 invalid scales	N/A	Replay N/A
Binary32-Z	24 valid	0 / 4 / 8 / 8	TF agreement 0.9674; FR agreement 0.7734
HA-FBMS	24 valid	0 / 2 / 4 / 5	TF agreement 0.9909; FR agreement 0.8880

Binary32-Z and HA-FBMS teacher-forced maximum KL values were 9.6921710968 and 0.007775790989; mean-case mean KL values were 0.06949804767 and 0.000709157912. Free-running maximum KL values were 30.496213913 and 28.262727737, but values after first divergence are path-dependent. The test is localized logical injection over a finite frozen target set and a 32-step short horizon, not physical HBM injection, a hardware fault-rate estimate, or evidence of universal per-target dominance.

S7 DDR mapping stability and source-aware generalization

The 32 DDR scenarios have weak positional agreement. Across 351 valid scenario pairs, mean and median Spearman correlations are 0.1311 and 0.0414. Mean within-source agreement is 0.3844, compared with 0.1039 across sources. This pattern indicates limited transfer of position ordering across the evaluated scenario families; it does not imply that individual DDR fault profiles are physically uniform.

The nested protocol uses five source-aware outer folds and grouped inner validation. Inner selection jointly evaluates shrinkage values of 0, 0.10, 0.25, 0.50, 0.75, and 1.0 and six frozen objectives. Complete shrinkage to uniform exposure with the mean objective is selected in all five outer folds. A fourfold multistart diagnostic yields no material calibration-objective improvement, so optimizer quality is not the dominant explanation for the generalization result.

Fold-level normalized ratios are sensitive to small identity-risk denominators. Thirteen of 216 scenario-by-representation components fall below the prespecified near-zero threshold, and the minimum identity risk is 6.714686851432498e-05. The reconstructed maximum component ratio is 2047.996093750181. The mean and median outer-fold ratios are 59.62921164892907 and 8.31474160853205, and the largest fold ratio is 256.8573982635019. These values are retained for transparency but are not headline effect sizes.

The extreme fold was reconstructed exactly from 48 held-out scenario-by-representation components. Six near-zero components contribute 255.99405924475042 to the fold mean of 256.8573982635019. P19 and P21 are not directly comparable: P19 pools exposure before forming representation-level ratios, whereas P21 forms scenario-level ratios before fold aggregation. The main text therefore retains the pooled P19 sensitivity metric and reports P21 as a source-aware distributional generalization audit.

Evidence item	Value	Interpretation
Pairwise Spearman	Mean 0.1311; median 0.0414	Weak cross-scenario position ordering
Source comparison	Within 0.3844; cross 0.1039	Source-conditioned structure does not transfer strongly
Nested selection	lambda = 1.0 in 5/5 folds	Position-specific weighting shrinks to uniform
Optimizer diagnostic	No material headroom	Search budget is not the dominant limitation
Denominator audit	13/216 near zero; minimum 6.71e-05	Mean ratios are denominator-sensitive
Fold ratios	Mean 59.6292; median 8.3147	Distribution retained; mean is not a headline effect size
Extreme reconstruction	Maximum component ratio 2047.9961	Traced to a small identity denominator

Table S5. DDR positional stability and denominator audit

S8 gfx942 runtime-permutation attribution

The P2D follow-up uses frozen REP_0025 at 4,194,304 groups with 30 warmup rounds, 15 paired timing blocks, and 100 launches per timed block. Identity, identity-mapped NOOP, real non-identity mapped, and PREPERMUTED variants all pass functional checks. PREPERMUTED applies the same frozen logical permutation before the timed kernel, and its normalization round trip is exact.

Median latencies are 1.024525032 ms for identity, 1.024526329 ms for NOOP, 1.332571640 ms for real mapped, and 1.023709145 ms for PREPERMUTED. Relative medians are -0.000328%, +30.074147%, and -0.078396% for NOOP, real mapped, and PREPERMUTED against identity. The real-mapped 95% interval is [+30.054129%, +30.088123%]. PREPERMUTED is near the identity scale, but its negative difference is not interpreted as acceleration.

Static ISA ratios are 1.301205 for total instructions and 1.454545 for VALU instructions. The mapped path adds AND and shift operations; the audited census contains no V_BFE, V_BFI, or V_PERM instruction and no scratch or LDS instruction. Dynamic ratios are 1.354430 for total executed instructions, 1.509091 for executed VALU instructions, 1.247829 for wave cycles, and 1.277183 for CU busy cycles.

VMEM instruction ratio is 1.000000. TCC fetch, hit, and miss ratios are 1.000004, 1.000033, and 1.000001. Scratch use is zero, LDS traffic is absent, and occupancy ratio is 0.976859 rather than a collapse. This negative evidence rules against memory traffic, cache behavior, spilling, LDS traffic, and occupancy collapse as the primary cause under the measured protocol.

The controlled A/B supports runtime non-identity permutation as the cause of the measured gfx942 kernel penalty. It does not assign an intrinsic 30% cost to HA-FBMS, establish an H800 cause, or measure end-to-end serving latency. Pre-normalization removes the timed-kernel penalty in this implementation, but is not claimed to be universally free.

Variant	Median ms	Relative median	Functional check
Identity	1.024525032	reference	pass
Identity NOOP	1.024526329	-0.000328%	pass
Real mapped	1.332571640	+30.074147%	pass
PREPERMUTED	1.023709145	-0.078396%	pass; round trip exact

Table S6. Controlled gfx942 causal A/B

S9 Implementation and archive provenance

Historical public artifact: v1.5.0-tjs-final. The frozen release is <https://github.com/psybao/MetaKV/releases/tag/v1.5.0-tjs-final> at commit e342a837b5bc39e593ac2106d3f1644880fc9ab6, archived at <https://doi.org/10.5281/zenodo.22750136>. This archive includes the CPU and Apple M4 portability evidence and its provenance; the frozen scientific numbers and earlier historical releases remain unchanged.

Table S7. Frozen deployment inputs

Artifact	SHA256
PIE reference script	cd6a48684214a8e3b453f3b8f32e7123fe9e689a265d96511aa9956b7ff18c61
Representation CSV	21c366e0b731152a8356673cbe4adfc03f5720010485ed1094f20ad8e1f5fb05
Mapping JSON	d7cd28781a259371419511e3ca5e74ba78358888596833574c13fb92c53da367

No additional supplementary figure is needed: manuscript Figures 1–6 already show the implementation path, representation geometry, characterization, deployment, model scale-up, and semantic response. Tables S1-S7 provide the supplementary numeric and provenance detail without creating new observations.

S10 Cross-ISA CPU and Apple M4 portability evidence

Article: MetaKV: Fault-Resilient Structured Metadata for Low-Precision LLM KV Caches

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Cross-ISA CPU codec microbenchmarks

Frozen experiments use the same 4,194,304-group workload, four representative configurations, 1/4/8 OpenMP threads, two warmup runs, 15 paired blocks, and 10,000 bootstrap resamples on Intel Core i7-14700KF, AMD Ryzen 7 7840HS, and Apple M4. The compiler-controlled x86-64 packages use clang 23.1.0; the Apple ARM64 package uses Homebrew clang 23.1.0 with libomp. Results are portability microbenchmarks, not GPU benchmarks or end-to-end LLM latency measurements.

Platform	ISA / OS	Compiler/runtime	Workload	Maximum absolute median difference (95% CI)	Validation
Intel Core i7-14700KF	x86-64 / Windows	LLVM-MinGW clang 23.1.0; OpenMP	4,194,304 groups; 4 reps; 1/4/8 threads	5.277% (0.766%, 10.290%)	12/12 within 8%; PASS
AMD Ryzen 7 7840HS	x86-64 / Windows	LLVM-MinGW clang 23.1.0; OpenMP	4,194,304 groups; 4 reps; 1/4/8 threads	6.303% (3.565%, 16.668%)	12/12 within 8%; PASS
Apple M4	ARM64 / macOS	Homebrew clang 23.1.0; libomp	4,194,304 groups; 4 reps; 1/4/8 threads	6.713% (5.946%, 8.087%)	12/12 within 8%; PASS

Apple M4 also passes separate functional checks on the PyTorch MPS and MLX Metal backends. MPS attention-like output is finite with maximum absolute error 5.96e-8 versus CPU. MLX preserves packed INT4 and metadata round trips bit-exactly, with zero maximum dequantization error in the tested sweep over sequence lengths 128, 512, 1024, and 2048. These checks make no performance, end-to-end LLM, physical-memory-mapping, or CPU-versus-GPU comparison claim.